Lecture VI - Computational Kindness

Programming with Python

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Key Concepts

Topics from the Lecture

- **Optimal Stopping**: How to decide **when to stop looking** for better options.
- **Explore/Exploit Tradeoff**: **Balancing** between trying new things and sticking with known options.
- **Caching**: Storing and reusing information by **sticking to a task** to improve performance.
- **Scheduling**: Efficiently managing **tasks and time.**
- **Randomness**: Understanding and working with **uncertainty**.

Kindness

Computational Kindness

Question: An idea what that is?

- Idea of introducing **empathy in algorithms**
- **Consider human cognitive load and limitations**
- Making choices that **reduce mental burden** for others
- Creating systems that are **easy to understand and interact**

Scheduling Meetings

- **Propose specific times!**
- Don't ask "when are you free?"
- **Reduces** cognitive load for others
- **Transforms** an open-ended problem

> ! Important
> **Now a simple yes/no decision!**

Communication

- **Be explicit rather than implicit**

- Provide **clear** options instead
- **Avoid** open-ended questions
- State your **preferences**

> ! **Important**
> **Avoid** deferring → "I'm fine with **whatever**"!

The Cost of Flexibility

- "I'm free whenever" **seems helpful**
- **But it is not!**
- Forces others to **consider all possibilities**
- **Increases** cognitive load
- Makes decision-making **more complex**

Principles

1. **Reduce Options**: Fewer choices lead to **better decisions**
2. **Be Explicit**: Clear constraints **help** others decide
3. **Consider Cognitive Load**: Design interactions minimizing **mental effort** for others
4. **Make Decisions**: **Taking responsibility** can be kind

Reflection

Think about it: How could you apply computational kindness in these scenarios?

- Planning a **group dinner** with friends
- Asking your professor for **thesis feedback**
- Coordinating a **team project** at work

How to continue?

How to continue learning?

- We have covered **a lot of topics**
- But there are **many more** to explore!

Bayes' Rule

- Fundamental theorem in probability theory
- Updates probability of hypothesis based on **new evidence**
- Used in **statistics, machine learning, and decision-making**

> ! **Important**
> Helps make **informed predictions and decisions** under uncertainty in the real world!

Overfitting

- Occurs when a model **learns the training data too well**
- **Captures noise and outliers** rather than pattern
- Results in **poor generalization to new data**

> ! **Important**
> Several ways to counter overfitting, e.g. cross-validation.

Game Theory

- Strategic interactions among **rational decision-makers**
- Analyzes situations with **multiple agents** and their strategies

> ! **Important**
> **Many applications** in economics, politics, and biology!

Networking

- Study of **information exchange** over networks
- Includes **understanding protocols and data transmission**
- Optimizes **network resources**

> ! **Important**
> **Important** in computer networks, the internet and social life!

Relaxation

- Simplifies **complex problems** to make them more tractable
- Involves **relaxing certain constraints**
- Provides insights or **approximate** solutions

Continue programming?

- The best way to continue learning is to **keep programming in the future**
- Potentially, you will continue to do so during your studies
- Coding in your Thesis is another great way to improve
- Try to find a way to apply programming in your work
- **There are many interesting topics to explore!**

Advent of Code

- Advent of Code is a fun way to keep programming

- Here you can solve programming puzzles during Advent
- It is **completely free** and ad-free and starts each year at 01.12.

That's it for the Lecture Series!

- We now have covered the basics of Python
- I hope you enjoyed the lecture and found it helpful
- If you have questions or feedback, **please let me know!**
- I wish you all the best for your studies and your career!

Q&A

Literature

Interesting literature to start

- Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.[1]

Books on Programming

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. Here
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

> **Note**
> Think Python is a great book to start with. It's available online for free. Schrödinger Programmiert Python is a great alternative for German students, as it is a very playful introduction to programming with lots of examples.

More Literature

For more interesting literature, take a look at the literature list of this course.

[1] The main inspiration for this lecture. Nils and I have read it and discussed it in depth, always wanting to translate it into a course.